

FIGHTER AND FIGHTER-BOMBER AIRCRAFT 1939–42

Although much faster and more manoeuvrable than their predecessors, the fighter aircraft of 1918 were little changed from those of 1914. During the 1930s, all-metal monoplane fighters came into their own (a rare exception being the Hawker Hurricane, which had a fabric covering), with vastly more powerful engines, the ability to mount additional machine-guns, and heavier “cannon”. By 1939, the most capable aircraft were reaching speeds in excess of 560kph (348mph), and could climb to 10,360m (34,000ft). Experts believed that such enhanced performance would put an end to the dogfights that were the essence of aerial combat during World War I, but they were soon proved wrong.

► SUPERMARINE SPITFIRE

Date	1939	Origin	UK
Wingspan	11.23m (36¾ft)		
Length	9.12m (30ft)		
Top speed	605kph (378mph)		
Engine	1,081.1kW (1,470hp) Rolls-Royce Merlin 45 in-line V-12		

The Spitfire's superb handling characteristics ensured its immediate success. Its continuous development – it was to be built in 13 main variants – kept it in service until well after the end of the war.

▼ MESSERSCHMITT BF 110

Date	1940	Origin	Germany
Wingspan	16.3m (53¾ft)		
Length	12.3m (40½ft)		
Top speed	560kph (350mph)		
Engine	Two 809kW (1,100hp) Daimler-Benz DB 601A in-line V-12s (inverted)		

Designed originally as a long-range bomber escort, the early Bf 110 displayed serious shortcomings in that role. Later, -F and -G variants were significantly more successful, particularly as radar-equipped night-fighters.



▲ MESSERSCHMITT BF 109

Date	1941	Origin	Germany
Wingspan	9.85m (32½ft)		
Length	8.65m (28½ft)		
Top speed	570kph (355mph)		
Engine	956.1kW (1,300hp) Daimler-Benz DB 601E in-line V-12 (inverted)		

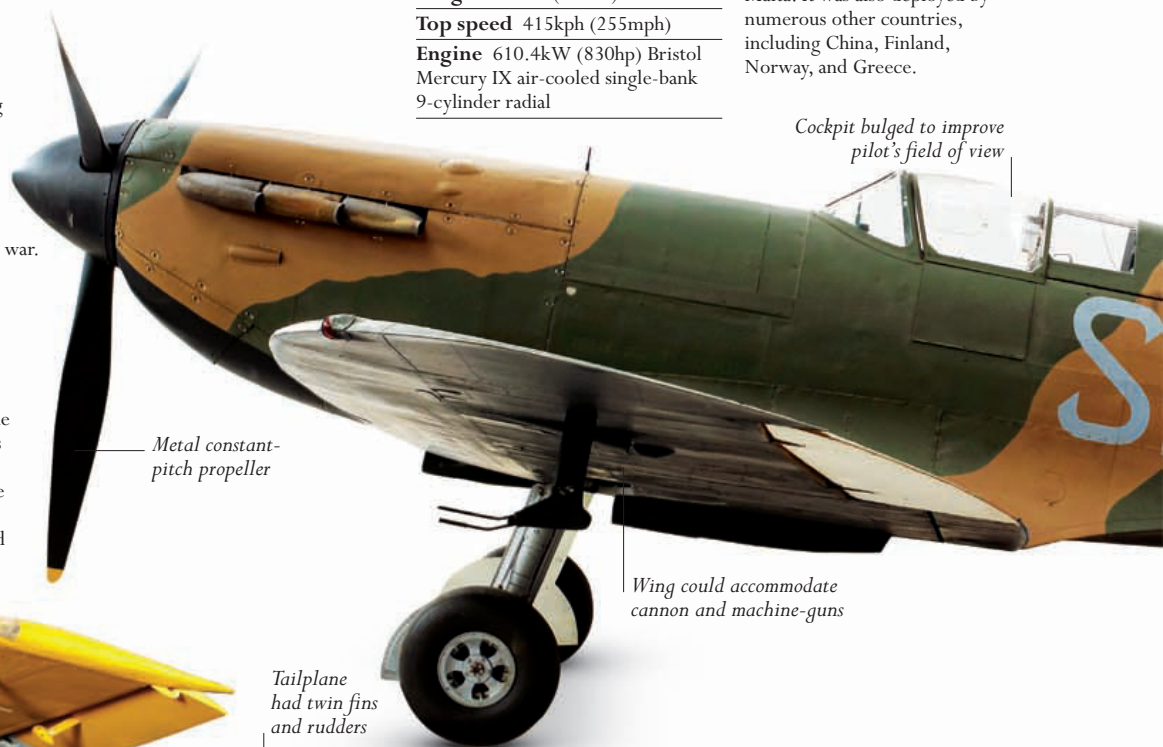
The Bf 109 beat stiff competition to become the *Luftwaffe's* first modern fighter aircraft. In all, some 35,000 were built in nine versions and many variants, with engines from Junkers or Daimler-Benz.



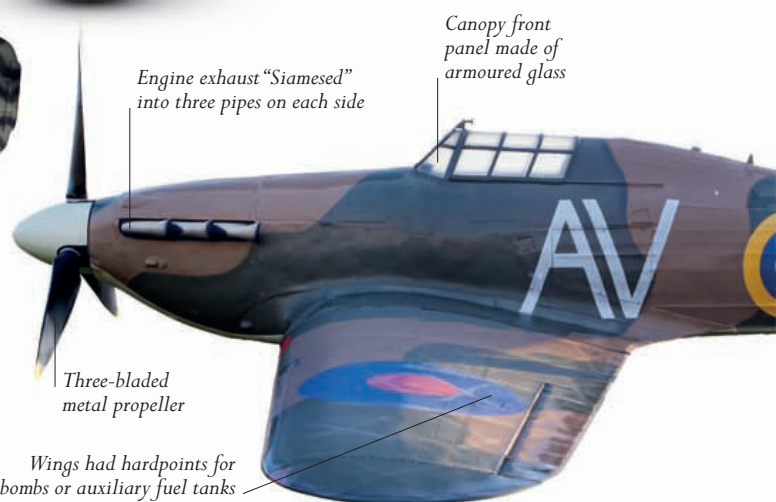
▲ GLOSTER GLADIATOR

Date	1937	Origin	UK
Wingspan	9.83m (32¼ft)		
Length	8.36m (27½in)		
Top speed	415kph (255mph)		
Engine	610.4kW (830hp) Bristol Mercury IX air-cooled single-bank 9-cylinder radial		

Although technically outdated, the Gladiator saw front-line service with the RAF in France, Norway, and (most famously) Malta. It was also deployed by numerous other countries, including China, Finland, Norway, and Greece.



Tailplane had twin fins and rudders



Engine exhaust “Siamesed” into three pipes on each side

Canopy front panel made of armoured glass

Three-bladed metal propeller

Wings had hardpoints for bombs or auxiliary fuel tanks

► FOCKE-WULF FW 190 WÜRGER

Date	1941	Origin	Germany
Wingspan	10.5m (34½ft)		
Length	8.84m (29ft)		
Top speed	655kph (410mph)		
Engine	1,132.6kW (1,540hp) BMW 801C2 air-cooled 2-bank 14-cylinder radial		

The most advanced fighter aircraft when it entered service, the Fw 190 remained unbeatable until confronted by the Spitfire Mk IX. An active development programme saw improved versions produced until 1945. The Fw 190 was the first important fighter of World War II to have a radial engine.

